DBNSR: Hyperspectral Image Super-Resolution Based on Dual-Branch Network

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***Abstract*—To address the problem that current hyperspectral image super-resolution methods struggle to effectively extract spatial–spectral information, resulting in degraded performance, this article proposes a dual-branch network for hyperspectral images super-resolution (DBNSR). In the proposed framework, one branch transforms hyperspectral image super-resolution into abundance-map super-resolution through unmixing, leveraging endmembers to represent spectral information and thereby enhancing spectral feature extraction. Simultaneously, the other branch performs super-resolution directly on the spatial domain of the original hyperspectral image to effectively extract spatial features. Furthermore, a dedicated high-frequency feature extraction (HFE) module is proposed to capture fine-grained details via spatial–spectral residual blocks (SSRBs), thereby enhancing super-resolution performance. Finally, a spatial–spectral feature fusion module is introduced to integrate the outputs of the dual branches, enabling effective fusion of spatial and spectral information. Experimental results on three datasets demonstrate that the proposed method outperforms current state-of-the-art methods both metrically and visually. The code is available at: https://github.com/fomanwong/DBNSR**

***Index Terms*—High-frequency feature extraction (HFE), remote sensing image, spectral unmixing, super-resolution.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain abundant spectral information, which helps improve the accuracy of tasks such as target detection and classification [1], [2], [3]. However, due to limitations in imaging conditions [4], [5], HSI typically have low spatial resolution, which restricts their further applications. Therefore, research on HSI super-resolution is of great significance.

HSI super-resolution techniques aim to enhance the spatial resolution of HSI. Existing HSI super-resolution methods are mainly divided into multi-image fusion-based super-resolution and single-image super-resolution. Multi-image fusion-based super-resolution methods [6], [7], [8], [9], [10] enhance the spatial resolution of HSI by fusing high-resolution multispectral or panchromatic images of the same scene with the super-resolved HSI. However, this approach requires strict alignment between the HSI and the high-spatial resolution images [11]. In contrast, single-image HSI super-resolution methods enhance the spatial resolution without requiring additional information, making them more valuable for practical applications. Therefore, this article focuses on single-image HSI super-resolution methods.

Single-image HSI super-resolution methods face greater challenges due to the lack of fused high-resolution image information. Traditional single-image HSI super-resolution methods include sparse regularization [12], [13], maximum a posterior estimation [14], [15], and low-rank approximation [16], [17], [18] methods. However, these traditional methods rely on handcrafted prior models and struggle to extract the rich information inherent in HSI, resulting in limited super-resolution performance.

Deep learning-based single-image HSI super-resolution methods leverage strong feature extraction capabilities to achieve better super-resolution results than traditional methods. Existing single-image HSI super-resolution networks can be divided into original HSI-based and abundance-map-based methods. Networks based on the original HSI extract deep spatial–spectral features to improve super-resolution accuracy. Considering that high-frequency spatial details are critical for improving super-resolution accuracy [19], [20], [21], early networks based on original HSI [22] employed 2-D convolutions for effective extraction of such information. However, 2-D convolution neglects high-dimensional spectral information, thereby limiting super-resolution performance. To address this, the authors in [23], [24], [25], [26], and [27] proposed 3-D convolution-based super-resolution methods, which improve the spectral reconstruction ability via convolutions along the spectral dimension. Nevertheless, these methods are computationally intensive, and 3-D convolution's uniform treatment of spatial–spectral features ignores their inherent physical differences, constraining super-resolution effectiveness. To enhance efficiency and spectral reconstruction, some researchers proposed grouping-based methods [28], [29], [30], [31] which boost model performance by learning localized spectral features. However, these methods tend to overlook spectral correlations between different groups, potentially leading to spectral discontinuity issues after groupwise reconstruction. Therefore, attention mechanisms [32] or intergroup information feedback mechanisms [33] were introduced to reconstruct intergroup information and enhance spectral reconstruction quality. To further enhance super-resolution, deeper networks for high-frequency feature extraction (HFE) or

Received 5 January 2026; revised 2 March 2026; accepted 24 March 2026. Date of publication 26 March 2026; date of current version 10 April 2026. *(Corresponding author: Yan Tian.)*

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Digital Object Identifier 10.1109/TGRS.2026.3678075

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transformer-based global attention mechanisms were adopted to improve high-frequency region reconstruction [34], [35], [36]. However, their high sensitivity to details and lack of physical constraints on spatial–spectral information may cause artifacts in low-frequency regions. Moreover, transformer-based methods require substantial memory and computational resources, compromising network efficiency. Abundance-map-based super-resolution methods [6], [37], [38] obtain mixed endmembers and corresponding abundance maps through unmixing, perform super-resolution on abundance maps, and then linearly mix them with endmembers to reconstruct high-resolution HSI. These methods impose strong physical constraints, reduce the coupling between spatial and spectral information by super-resolving the abundance maps, and avoid spectral distortion commonly caused by spatial super-resolution. Simultaneously, the endmember inversion can effectively learn and represent spectral information, improving spectral reconstruction capability [37]. However, abundance inversion in complex mixed pixels (e.g., high-frequency regions) may introduce errors [39], [40], [41]. These errors smooth high-frequency regions, cause loss of spatial details, and ultimately degrade super-resolution performance.

In summary, existing single-image super-resolution methods still struggle to effectively extract spatial–spectral information from HSI.

1) Original HSI-based super-resolution networks improve accuracy in high-frequency regions by using deep HFE or global attention mechanisms but tend to produce artifacts in low-frequency regions with spectral errors, degrading overall super-resolution performance.
2) Abundance-map-based super-resolution networks improve spectral reconstruction performance through endmember inversion of the unmixing process, but inversion errors in abundance maps cause spatial information loss, reducing spatial super-resolution accuracy.

To address these issues, this article proposes a dual-branch network for hyperspectral images super-resolution (DBNSR) combining original HSI super-resolution (referred to as the original-image branch) and abundance-map super-resolution (referred to as the abundance-map branch). First, the abundance-map branch serves as the main branch, effectively extracting spectral information via endmember inversion. Meanwhile, the original-image branch enhances the spatial information of the main branch through deep spatial detail extraction, enabling more effective spatial feature extraction. In addition, by leveraging the land cover distribution information from the abundance map, the spectral consistency of homogeneous regions is improved, helping to reduce artifacts and enhance the super-resolution performance. Second, a dedicated HFE module is proposed, which extracts high-frequency details of HSI through spatial–spectral residual blocks (SSRBs), improving super-resolution accuracy. Finally, a spatial–spectral feature fusion (SSF) module is introduced to effectively fuse results from the dual-branch network, further enhancing the model performance.

The overall processing flow of the network is illustrated in Fig. 1. In the abundance-map branch, the input low-resolution HSI is first unmixed to obtain endmembers and their corresponding abundance maps. The abundance maps are then super-resolved to obtain high-resolution abundance maps, which are subsequently linearly mixed with the endmembers to produce the high-resolution HSI^1. Meanwhile, the original-image branch directly performs super-resolution on the low-resolution HSI to generate the high-resolution HSI^2. Then, the SSF module of the network combines HSI^1 and HSI^2 to produce the final super-resolution result. Notably, the super-resolution processes in both branches share modules such as HFE and upsampling, enabling comprehensive spatial information learning and further enhancing the network's performance. Therefore, the main contributions of this article are as follows.

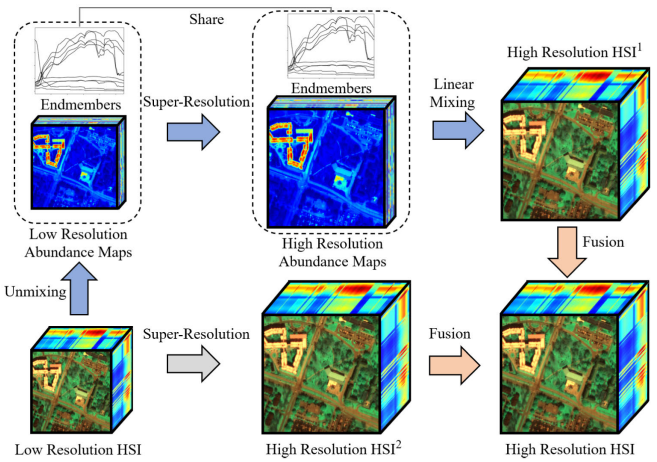


Fig. 1. Procedure of the proposed DBNSR. In the abundance-map-based branch, the low-resolution HSI is decomposed into abundance maps and endmembers. The abundance maps are super-resolved and mixed with the endmembers to generate the high-resolution HSI^1. In the original HSI-based branch, high-resolution HSI^2 is directly generated from the low-resolution HSI via a super-resolution network. Finally, HSI^1 and HSI^2 from the two branches are fused to produce the final high-resolution HSI.

1) A dual-branch super-resolution network is proposed that combines abundance map and original-image branches to enhance the network's spatial–spectral information extraction ability, thereby improving HSI super-resolution performance.
2) A dedicated HFE module is introduced that extracts high-frequency details through SSRBs, enhancing feature extraction performance.
3) An SSF module is proposed to effectively fuse the dual-branch super-resolution results, improving the model's super-resolution performance.

II. Related Works

The proposed DBNSR in this article is a group-based single-image HSI super-resolution network, which employs an HFE method as its core feature extraction module. This section provides a concise introduction to such networks and their feature extraction strategies.

A. Group-Based Super-Resolution

Compared with traditional three-channel RGB images, HSI contains rich spectral information. Existing image super-

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resolution techniques are primarily designed for RGB images [42], [43], [44], [45], and directly applying these methods to HSI often leads to suboptimal performance due to the neglect of spectral dimensional characteristics. Therefore, current research on HSI super-resolution mainly focuses on enhancing the ability to extract spectral information from HSI. While simple methods using 3-D convolution can improve spectral feature extraction, they suffer from low computational efficiency. Moreover, the indiscriminate treatment of spatial and spectral information in such methods ignores their physical differences, limiting the overall super-resolution performance. To address these issues, researchers have proposed a group-based super-resolution strategy, which first divides the HSI into multiple spectral groups and performs super-resolution reconstruction on all grouped images through shared network weights. The reconstructed results from each group are then recombined and fused according to their spectral band relationships to obtain the final full-band hyperspectral super-resolution result. By adopting this grouping approach, the processing of high-dimensional HSI is decomposed into multiple lower dimensional subtasks, effectively reducing model complexity while better exploiting spectral information. This strategy enhances the spectral fidelity and improves the overall super-resolution performance. For example, Jiang et al. [34] proposed SSPSR, which performs two-stage upsampling: spectral group-based super-resolution in the first stage and entire spectral super-resolution in the second, significantly improving the reconstruction performance. However, spectral grouping may overlook correlations between groups, affecting the continuity of spectral reconstruction. To mitigate this, Wang et al. [33] introduced RFSR, a recurrent feedback super-resolution network that employs intergroup information feedback and spatial–spectral regularization to enhance spectral reconstruction. Similarly, Liu et al. [32] proposed the SGARDN model, which computes attention across groups using feedback information to guide spectral fusion. SFCSR [31] improves the quality of spectral reconstruction by increasing the number of groups to enhance the proportion of overlapping spectra between groups. Therefore, it can be concluded that enhancing the ability to extract spatial–spectral information is currently the primary challenge in HSI super-resolution research.

B. High-Frequency-Based Super-Resolution

Residual learning is a widely adopted structure in image super-resolution networks, as residual connections facilitate the extraction of high-frequency features, thereby benefiting detail reconstruction in super-resolution images. As a result, many super-resolution studies have focused on improving HFE. Lim et al. [19] proposed an effective residual block (RB) for single-image super-resolution, which has since been widely applied [46], [47], [48], [49]. Building on this, Du et al. [50] introduced an enhanced RB (ERB), which strengthens HFE through nonlinear activation and cross-residual connections. To further boost super-resolution performance, many researchers [21], [51] have incorporated channel and spatial attention mechanisms to enhance the network's focus on critical bands and pixels, thereby improving super-resolution performance. For instance, average pooled spectral features are commonly used to compute spectral attention [34]. In addition, second-order attention networks SAN [52] and holistic attention networks HAN [53] have been proposed to better capture important spatial regions. In recent years, increasing attention has been paid to capturing nonlocal dependencies within HFE modules [54], [55], [56], [57]. By computing long-range correlations, these models can better attend to critical bands and pixels. For example, Hu et al. [37] leveraged transformers to extract long-range dependencies between abundance maps, enhancing the model's attention to key abundances and corresponding endmembers. Chen et al. [35] proposed a cross-domain spatial–spectral network CST that captures long-range spatial–spectral dependencies via window-based transformers, improving super-resolution accuracy. However, extracting nonlocal relations imposes high demands on memory and computation. To address this, researchers have attempted lightweight designs to improve efficiency, but these often come at the cost of reduced performance. Furthermore, the lack of physical constraints in spatial–spectral relationships and sensitivity to high-frequency details can lead to artifacts and lower spectral reconstruction accuracy. Thus, there is a need for a dedicated HFE method to further enhance super-resolution performance.

III. Proposed DBNSR

A. DBNSR Framework

From the previous analysis, it can be seen that current HSI super-resolution methods mainly face two problems.

1) Super-resolution networks based on the original HSI improve super-resolution performance through high-frequency detail extraction. However, these methods lack physical constraints related to spatial–spectral information. Enhancing spatial high-frequency detail extraction may lead to artifacts and result in poor spectral reconstruction performance.
2) Super-resolution networks based on abundance maps possess strong physical constraints, which help to reduce the coupling between spatial and spectral information and improve the extraction of spectral information. However, this approach relies on the accuracy of abundance unmixing of mixed pixels. In regions with complex mixed pixels and rich high-frequency details, abundance estimates are prone to deviation [39], [40], [41], thereby weakening the reconstruction performance in these regions.

To leverage the strengths of both networks, this article proposes DBNSR, a dual-branch parallel super-resolution network built upon the original image and the abundance map. This parallel design effectively avoids the error-propagation issues commonly observed in sequential architectures. If abundance-map super-resolution is performed first, inversion errors may introduce structural distortions in high-frequency regions; if original-image super-resolution is applied first, spectral artifacts may occur, undermining the accuracy of subsequent spectral reconstruction. In addition, the parallel structure ensures that the fusion module receives clean

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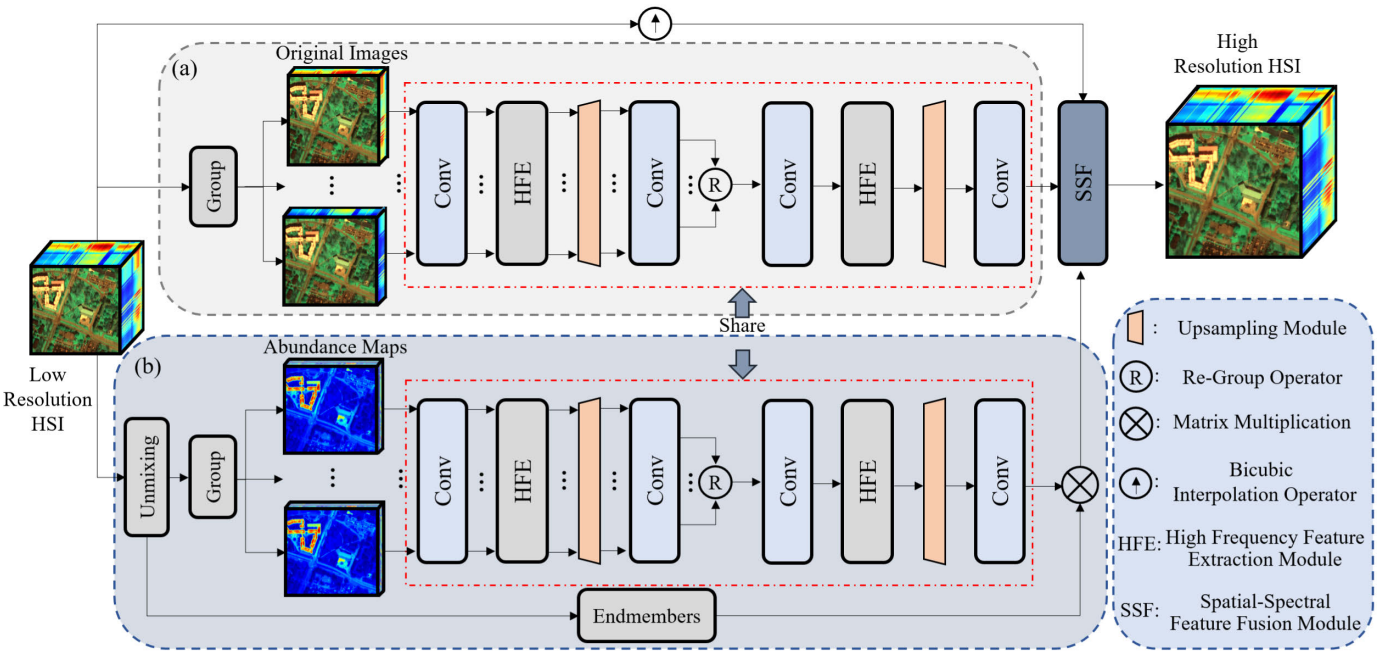


Fig. 2. DBNSR network architecture. (a) Super-resolution network branch based on the original HSI. (b) Super-resolution network branch based on the abundance maps.

features originating from different information domains, preventing premature information mixing and feature contamination that often happens in sequential designs. This helps maintain the independence of each branch and maximally preserves their respective advantages. Consequently, the network can achieve complementary and optimally integrated spatial–spectral information without being affected by intermediate error interference.

Therefore, in this article, the proposed DBNSR adopts the abundance-map branch as the main branch, leveraging physical constraints to ensure the reconstruction effectiveness from the spectral dimension. The original-image branch supplements spatial information for the main branch by extracting high-frequency details from the original image, thereby achieving effective spatial information extraction. Specifically, DBNSR integrates the spatial information of both the original image and the abundance map. Through shared learning between the two branches, the model effectively enhances its spatial reconstruction capability, thus improving super-resolution accuracy in high-frequency detail regions. Meanwhile, spectral information is effectively extracted via endmember inversion from the abundance-map branch. Combined with the spatial distribution of land cover in the abundance map, the spectral consistency of homogeneous regions is improved, which helps reduce the occurrence of artifacts. In addition, considering that existing HFE methods mostly use lightweight attention mechanisms or primarily focus on spectral feature learning, their ability to improve the network performance is limited. This article proposes a dedicated HFE module, which utilizes SSRBs and a streamlined spectral feature extraction structure to more effectively model high-frequency details in HSI. Finally, considering that existing feature fusion strategies often overlook the potential complementarity between features, which affects super-resolution performance, this article proposes a spatial–spectral fusion module (SSF). During fusion, the output of the abundance-map branch is preserved, while the output of the original-image branch is transformed through mapping operations. This helps to maintain the spectral consistency in homogeneous regions and promotes effective fusion of spatial details, thereby improving the overall super-resolution performance of the network.

The overall architecture of the DBNSR network is shown in Fig. 2. The network includes an abundance-map-based super-resolution branch (abundance-map branch), an original HSI-based super-resolution branch (original-image branch), an HFE module, and a spatial–spectral fusion (SSF) module. The basic processing flow of the network is as follows: first, the low-resolution HSI input is processed through the abundance branch and the original-image branch to obtain two high-spatial-resolution HSI outputs. Then, the SSF module integrates the outputs from both branches to generate the final super-resolution HSI result. In the dual-branch network's super-resolution process, HFE and upsampling modules are shared to facilitate comprehensive spatial information learning and efficient feature extraction.

1) Original-Image Branch: According to existing research [32], [33], [34], group-based learning reduces model complexity and helps the network to better learn the spatial details of HSI. Moreover, performing super-resolution in two stages—first grouped spectral image super-resolution and then on the entire spectral image super-resolution—helps the network to learn spectral correlations between groups, thereby improving HSI reconstruction quality. Therefore, this article adopts a staged progressive super-resolution strategy that follows a group-then-entire approach.

The structure of the original-image branch network is shown in Fig. 2(a). In the first stage of super-resolution, the input low-resolution HSI is initially grouped spectrally

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for preliminary shallow feature extraction. Subsequently, each group undergoes detailed feature extraction through the HFE module, followed by upsampling to obtain medium-resolution features. Finally, spectral feature mapping and recombination are performed across all bands in preparation for the second stage of super-resolution. In the second stage, the network similarly conducts shallow feature extraction and HFE via the HFE module, followed by a second round of upsampling and feature mapping to obtain the final high-resolution features. The specific process is as follows:

a) Grouped HSI super-resolution: The network first divides the input low-resolution HSI data $x_{LR} \in \mathbb{R}^{h\times w\times L}$ into groups according to spectral order: $\{x_{LR}^1, x_{LR}^2, \ldots, x_{LR}^G\}$. Here, $h \times w$ denotes the spatial resolution, L is the number of spectral bands, and G is the number of groups, and each group has a spectral band length of l(empirically set to 16) with an overlap of two bands between adjacent groups. For each group x_{LR}^g, shallow feature extraction [34] is performed to obtain $sF_{LR} \in \mathbb{R}^{h\times w\times dim}$

$$sF_{LR}^g = f_{SF}^1\left(x_{LR}^g\right) \tag{1}$$

where $f_{SF}^1(\cdot)$ is the shallow feature extraction module in the first stage, dim is the number of feature channels (empirically set to 180), and $g \in \{1, 2, \ldots, G\}$. Then, HFE is applied to each group sF_{LR}^g, producing features F_{LR}^g

$$F_{LR}^g = f_{HFE}^1\left(sF_{LR}^g\right) \tag{2}$$

where $f_{HFE}^1(\cdot)$ is the high-frequency feature extraction module in the first stage. Next, the features F_{LR}^g are upsampled to obtain medium-resolution features $F_{MR}^g \in \mathbb{R}^{hd\times wd\times dim}$

$$F_{MR}^g = f_{UP}^1\left(F_{LR}^g\right) \tag{3}$$

where d and $f_{UP}^1(\cdot)$ denote the upscaling factor and upsampling module of the first stage, respectively. Then, the medium-resolution grouped features undergo "spectral reconstruction" [34] through the feature channel mapping, generating the reconstructed feature $rF_{MR}^g \in \mathbb{R}^{hd\times wd\times l}$

$$rF_{MR}^g = f_{REC}^1\left(F_{MR}^g\right). \tag{4}$$

Finally, the medium-resolution grouped features are spectrally regrouped to obtain the complete band information feature $F_{MR} \in \mathbb{R}^{hd\times wd\times L}$

$$F_{MR} = \left[rF_{MR}^1, rF_{MR}^2, \ldots, rF_{MR}^G\right] \tag{5}$$

where $[\cdot]$ represents the regrouping operation.

b) Entire HSI super-resolution: The input feature F_{MR} undergoes shallow feature extraction and HFE to obtain $F'_{\mathrm{MR}} \in \mathbb{R}^{hd\times wd\times dim}$

$$F'_{MR} = f_{HFE}^2\left(f_{SF}^2\left(F_{MR}\right)\right) \tag{6}$$

where $f_{SF}^2(\cdot)$ and $f_{HFE}^2(\cdot)$ denote the shallow feature extraction module and HFE module in the second stage, respectively. Then, F'_{MR} undergoes a second upsampling operation and channel mapping to obtain the high-resolution features $F_{HR}^x \in \mathbb{R}^{H\times W\times L}$ from the original-image branch

$$F_{HR}^x = f_{REC}^2\left(f_{UP}^2\left(F'_{MR}\right)\right) \tag{7}$$

where $f_{REC}^2(\cdot)$ and $f_{UP}^2(\cdot)$ denote the reconstruction mapping and upsampling module in the second stage, respectively, and $H \times W$ is the target high-resolution size.

2) Abundance-Map Branch: The abundance-map-based super-resolution network leverages the unmixing-based learning of abundances and endmembers to better capture spectral information. Considering that fully convolutional networks help preserve spatial information [37], this work adopts a fully convolutional network as the abundance extraction module. Additionally, 1×1 convolution kernels are used to model the endmembers for linearly combining with the abundances to reconstruct the HSI. To enhance performance, the abundance map super-resolution branch shares both high-frequency extraction and upsampling modules with the original-image branch, thereby leveraging the rich spatial information from both the HSI and abundance maps to improve feature extraction capability.

The structure of the abundance-map branch is illustrated in Fig. 2(b). The network first obtains abundance maps and endmembers through an unmixing subnet. It then performs first-stage grouped abundance-map super-resolution with HFE, followed by second-stage entire abundance-map super-resolution to generate high-resolution abundance maps. Finally, the high-resolution HSI is reconstructed via a linear combination of these abundance maps with the endmembers. The process is as follows.

a) Unmixing: Given an input HSI $x_{LR} \in \mathbb{R}^{h\times w\times L}$, the network first uses a CNN to estimate the abundance map $a \in \mathbb{R}^{h\times w\times C}$

$$a = f_{CNN}\left(x_{LR}\right) \tag{8}$$

where C is the number of endmembers (empirically set to 30), and $f_{CNN}(\cdot)$ denotes the convolutional module. The abundance a satisfies the nonnegativity (ANC) and sum-to-one (ASC) constraints, enforced via ReLU and Softmax, respectively. The endmembers are modeled as 1×1 convolution kernels $e \in \mathbb{R}^{1\times 1\times C\times L}$, and remain physically plausible under reconstruction and abundance constraints [58], [59]. A point-wise convolution between the abundance maps and endmembers reconstructs the low-resolution HSI $x_r \in \mathbb{R}^{h\times w\times L}$

$$x_r = a * e \equiv ae \tag{9}$$

where $\equiv$ indicates that the pointwise convolution is equivalent to a linear combination of abundance maps and endmembers.

To ensure the reliability of unmixing, the unmixing module here first adopts a joint optimization mechanism based on multi-resolution reconstruction, achieving collaborative enhancement through dual-resolution reconstruction losses. At the low-resolution level, the abundance maps and endmembers are used to reconstruct the input HSI, ensuring reconstruction accuracy. At the high-resolution level, the same set of endmembers is combined with the super-resolved abundance maps to generate the target HSI. This cross-resolution constraint strengthens the model's generalization ability and helps prevent overfitting. Second, a moderately relaxed design is applied to the selection of the number of endmembers, tailored for super-resolution tasks. Increasing the number of endmembers appropriately enhances the model's nonlinear representation capability. The additional virtual endmembers can capture highly mixed pixels in high-frequency detail regions, effectively mitigating the loss of spatial details.

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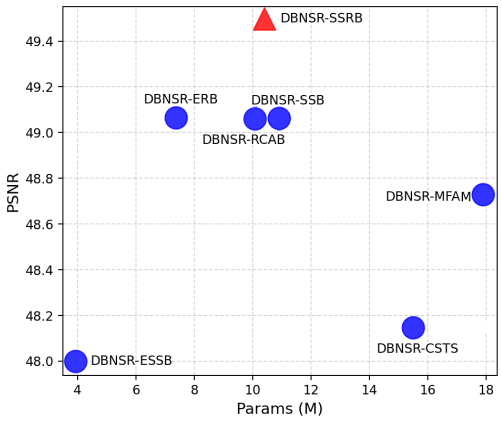


Fig. 3. Comparison of super-resolution performance (PSNR) and parameter size (M, input: $48 \times 16 \times 16$) under the DBNSR framework using different feature extraction modules on the Houston dataset with a $\times 4$ scale factor. ERB is the basic feature extraction module in FMEN [50], RCAB in RCAN [21], SSB in SSPSR [34], MFAM in CETNet [49], ESSB in EUNet [28], and CSTS in CST [35].

b) Abundance-Map super-resolution: The super-resolution process in the abundance-map branch is similar to that of the original-image branch, consisting of a two-stage approach: the first stage performs super-resolution on grouped abundance maps, and the second stage processes the entire abundance map. Specifically, the abundance maps with C channels are first projected to L channels for shared processing. Then, the projected abundance maps are grouped as $\{a^1, a^2, \ldots, a^G\}$, and passed through the following pipeline to obtain the final high-resolution features F_{HR}:

$$\begin{aligned} F^a_{MR} &= \left[\ldots, f^1_{REC}\left(f^1_{UP}\left(f^1_{HFE}\left(f^1_{SF}\left(a^g\right)\right)\right)\right), \ldots\right] \\ F_{HR} &= f^2_{REC}\left(f^2_{UP}\left(f^2_{HFE}\left(f^2_{SF}\left(F^a_{MR}\right)\right)\right)\right) \end{aligned} \tag{10}$$

where $f^1_{SF}(\cdot)$, $f^1_{HFE}(\cdot)$, $f^1_{UP}(\cdot)$, $f^1_{REC}(\cdot)$, f^2_{SF}, $f^2_{HFE}(\cdot)$, $f^2_{UP}(\cdot)$, and $f^2_{REC}(\cdot)$ are shared with the processing modules of the original-image branch. The symbol $[\cdot]$ represents the regrouping of the features across all abundance groups.

c) Linear mixing: The high-resolution abundance map F_{HR} is mapped to C channels and linearly mixed with the endmembers to obtain the high-resolution features of the abundance branch $F^a_{HR} \in \mathbb{R}^{H \times W \times L}$

$$F^a_{HR} = F_{HR} * e \equiv F_{HR} e. \tag{11}$$

B. HFE Module

Compared with traditional nonlearning methods, learning-based HFE networks possess a much stronger capacity for spatial–spectral feature modeling. In recent years, feature extraction structures based on high-frequency RBs and attention mechanisms have been extensively studied [19], [46], [47], [48]. However, existing approaches still suffer from two major limitations: 1) although attention mechanisms enhance feature selectivity, they introduce high computational and memory overhead, making it difficult for lightweight designs to balance performance and efficiency and 2) high-frequency RBs typically emphasize spectral feature extraction, resulting in insufficient spatial information modeling.

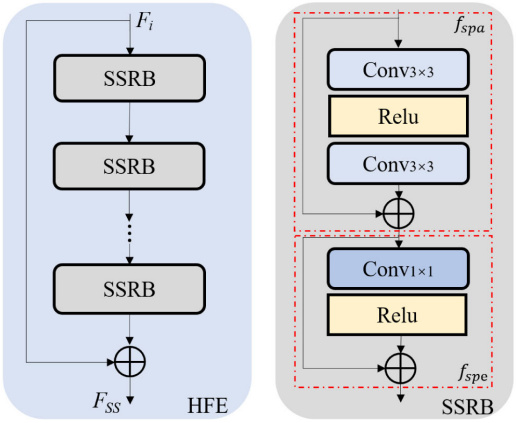


Fig. 4. HFE module and its basic components.

To address these issues, we propose a dedicated HFE module, which is constructed by cascading multilevel SSRBs. Unlike conventional RBs, SSRBs are structurally reorganized and functionally specialized under the DBNSR framework. First, at the module level, SSRB adopts a "spatial-dominant and spectrally lightweight" convolutional configuration. By adjusting the proportion and ordering of 3×3 and 1×1 convolutions, the module strengthens spatial HFE while reducing redundant spectral coupling, achieving a balance between performance and efficiency. Second, at the framework level, SSRB works collaboratively with the abundance-map branch: the former focuses on enhancing spatial details, while the latter ensures accurate spectral reconstruction. This enables task-oriented spatial–spectral decoupling and joint optimization. As shown in Fig. 3, compared with existing feature extraction modules, SSRB achieves the best balance between parameter size and performance.

As shown in Fig. 4, the HFE module first uses a sequence of SSRB as the backbone to deeply extract the high-frequency features of the HSI. In each SSRB, a 3×3 convolution is used to extract spatial details from the input, followed by a nonlinear activation to enhance feature representation. Then, a 1×1 convolution is applied for spectral feature extraction. Residual connections are employed throughout the spatial and spectral extraction stages to enable effective learning of high-frequency details, thereby enhancing the network's ability to recover fine structures. The detailed process is as follows.

1) Spatial Detail Extraction: Given an input feature F_i, spatial information is extracted through a 3×3 convolution and residual connection to obtain F_{spa}

$$F_{spa} = F_i + f_{spa}\left(F_i\right) \tag{12}$$

where $f_{spa}(\cdot)$ denotes the spatial feature extraction module, which consists of two 3×3 convolution layers with an *ReLU* activation in between.

2) Spectral Detail Extraction: The spatially enhanced feature F_{spa} is then processed using a 1×1 convolution and residual connection to extract spectral details, yielding F_{spe}

$$F_{spe} = F_{spa} + f_{spe}\left(F_{spa}\right) \tag{13}$$

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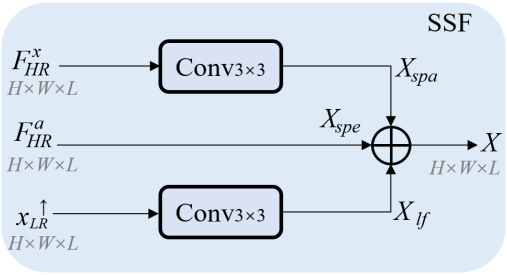


Fig. 5. SSF module.

where $f_{spe}(\cdot)$ represents the spectral feature extraction module, consisting of a single 1×1 convolution followed by an *ReLU* activation.

3) Deep Spatial–Spectral Detail Extraction: By cascading multiple SSRBs along with a global skip connection, the final spatial–spectral feature F_{ss} is obtained

$$F_{ss} = F_i + f_{SSRB}^{n}\left(f_{SSRB}^{(n-1)}\left(\ldots f_{SSRB}^{1}\left(F_i\right)\ldots\right)\right) \tag{14}$$

where $f_{SSRB}(\cdot)$ denotes the spatial–spectral RB, and n is the number of cascaded SSRB, which is set to 6 in our implementation.

C. SSF Module

The output of the original-image branch contains rich spatial details, whereas the abundance-map branch provides higher spectral reconstruction accuracy. How to effectively merge these two types of information in the spatial–spectral domain is key to improving the overall reconstruction quality. Existing super-resolution networks [34] typically add spatial and spectral features directly, followed by a convolutional smoothing layer to enhance fusion quality. In study [49], multiple convolutional and nonlinear layers are further applied after the direct addition of spatial–spectral features to strengthen the fusion capability. However, such "direct addition" strategies often overlook the inherent complementarity between spatial and spectral features (e.g., such as structure and detail), potentially leading to detail loss or spectral distortion. In study [37], a differential-learning strategy is adopted, where the differential information between spatial–spectral features and low-frequency components is separately mapped using 3×3 and 1×1 convolutions to improve the reconstruction accuracy. Nevertheless, such differential strategies tend to emphasize feature similarity and weaken complementary expressions between features, limiting the extraction of informative spatial–spectral details.

To address these challenges, this article proposes a spatial–spectral fusion (SSF) module, whose structure is illustrated in Fig. 5. The core design principle of SSF is to use the spectrally reliable output of the abundance branch as the fusion baseline and subsequently conduct guided spatial detail enhancement. Specifically, SSF first takes the abundance-branch output as the foundation to preserve spectral fidelity and maintain consistency within homogeneous regions. Then, the output of the original-image branch together with

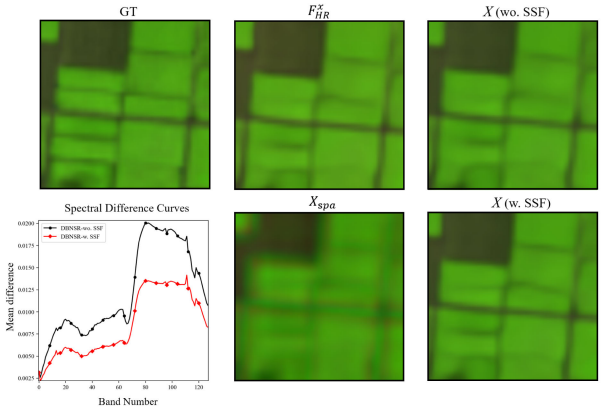


Fig. 6. Comparison of super-resolution reconstruction results, GT, and mean spectral difference curves of DBNSR with (w.) and without (wo.) the SSF module on the Chikusei test set with a ×4 super-resolution scaling factor. For visualization, the images use bands 70, 100, and 36 as the R, G, and B channels, respectively.

the upsampled input is fed into a lightweight convolutional mapping layer to selectively inject spatial detail information. This design enables structured, complementarity-driven spatial–spectral fusion. Accordingly, the fusion module mainly consists of the spatial-feature mapping of the original-image branch output and the low-frequency spatial-feature mapping of the original input.

1) Spatial Feature Mapping: The output F_{HR}^{x} of the original-image branch contains rich spatial information. Mapping this output helps to better integrate spatial details. The mapping can be defined as follows:

$$X_{spa} = f_{SFM}\left(F_{HR}^{x}\right) \tag{15}$$

where $f_{SFM}(\cdot)$ represents the convolutional mapping function for the original-image branch and X_{spa} is the resulting mapped spatial feature.

2) Low-Frequency Spatial Feature Mapping: The original low-resolution input x_{LR} primarily contains low-frequency information. Incorporating it into the fusion helps to maintain the stability of the spatial–spectral structure. The mapping is defined as follows:

$$X_{lf} = f_{LFM}\left(f_B\left(x_{LR}\right)\right) \tag{16}$$

where $f_{LFM}(\cdot)$ denotes the convolutional mapping of the input HSI, $f_B(\cdot)$ is the bicubic upsampling module, and X_{lf} is the resulting mapped low-frequency spatial feature. Finally, the above mapping results undergo final fusion

$$X = X_{spe} + X_{spa} + X_{lf} \tag{17}$$

where $X_{spe} = F_{HR}^{a}$. Fig. 6 illustrates visualizations of super-resolution results and intermediate features with and without the SSF module. It can be observed that direct feature addition (without SSF) tends to overlook the complementary advantages of spatial and spectral features, thereby reducing the overall reconstruction quality. With the inclusion of the SSF module, spectral reconstruction errors are significantly reduced, while the spatial edges extracted from the original-image branch are further enhanced. As a result, the final super-resolved output exhibits more refined structural details.

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These observations demonstrate that SSF effectively leverages the complementary strengths of the two branches, leading to improved super-resolution performance.

D. Loss Function

The training loss of the DBNSR network includes HSI super-resolution reconstruction loss and HSI unmixing reconstruction loss. According to the existing research [28], [34], [37], this article adopts L_1 loss and spectral angle mapping loss L_{sam} to measure the network's loss. The L_1 loss is used to measure absolute loss between pixels. Compared to L_2 loss, L_1 loss can produce a more balanced error distribution, beneficial for reconstructing rich details. Its expression is

$$\mathcal{L}_1(\Theta) = \frac{1}{N}\sum_{i=1}^{N}\left\|X_1^i - X_2^i\right\|_1 \tag{18}$$

where X_1 and X_2 represent the reconstructed image and corresponding ground truth (GT), respectively, Θ denotes the network parameters, and N is the total number of pixels.

The L_{sam} loss is used to measure spectral consistency of the reconstructed image, aiding in the restoration of spectral information after HSI reconstruction. Its expression is

$$\mathcal{L}_{sam}(\Theta) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\pi}\arccos\left(\frac{X_1^i \cdot X_2^i}{\left\|X_1^i\right\|_2 \cdot \left\|X_2^i\right\|_2}\right). \tag{19}$$

Therefore, this article uses L_1 and L_{sam} to measure the spatial and spectral dimension losses of the low-resolution HSI reconstruction x from unmixing and the high-resolution HSI reconstruction X from super-resolution

$$\mathcal{L}_{DBNSR} = \mathcal{L}_{spa} + \lambda\mathcal{L}_{spe} \tag{20}$$

where λ is the loss mixing weight, empirically set to 0.1. L_{spa} expression is

$$\mathcal{L}_{spa} = \mathcal{L}_1(x, \hat{x}) + \mathcal{L}_1\left(X, \hat{X}\right) \tag{21}$$

where $\hat{x}$ and $\hat{X}$ denote the low-resolution HSI and high-resolution HSI GT, respectively. L_{spe} expression is

$$\mathcal{L}_{spe} = \mathcal{L}_{sam}(x, \hat{x}) + \mathcal{L}_{sam}\left(X, \hat{X}\right). \tag{22}$$

By training with $\mathcal{L}_{DBNSR}$ loss, learning of unmixing reconstruction loss helps the model better explore spectral information of HSI, combined with super-resolution reconstruction loss to enhance recovery of spatial details.

IV. Experiments

In this section, we first introduce the experimental datasets and implementation details. We then compare the proposed DBNSR with several state-of-the-art HSI super-resolution methods. Finally, we conduct ablation studies to validate the effectiveness of each component in DBNSR.

A. Datasets and Implementation Details

1) Datasets: To validate the effectiveness of the proposed DBNSR, we compare it with state-of-the-art methods on three real-world HSI datasets: Chikusei [60], Houston,[1] and Pavia [61]. The Chikusei dataset was captured using the Headwall Hyperspec-VNIR-C imaging sensor. It contains 128 spectral bands covering the 363–1018 nm range, with a spatial resolution of 2517×2335. The Houston dataset (part of the 2018 IEEE GRSS Data Fusion Contest) was acquired by the ITRES CASI-1500 hyperspectral imager. It consists of 48 spectral bands ranging from 380 to 1050 nm and has a spatial resolution of 4172×1202. The Pavia dataset was acquired using the reflective optics system imaging spectrometer (ROSIS) sensor. It includes 103 bands spanning 430–860 nm, with a spatial resolution of 1096×1096.

2) Implementation Details: This article selects seven state-of-the-art single-image HSI super-resolution methods for comparison, including SSPSR [34] (TCI'20), GDRRN [29] (BigMM'18), SFCSR [31] (TIE'21), EUNet [28](IF'23), RFSR [33] (TGRS'22), SNLSR [37] (TIP'24), and CST [35] (TIP'24). All methods are trained and tested on the same datasets, and their convolutional layer channel numbers and other parameters are adjusted according to the original papers and the specific HSI datasets to ensure optimal performance. In the proposed DBNSR method, the number of endmembers in the abundance-map super-resolution network is empirically set to c = 30; the number of cascaded HFE modules is set to 6; the number of feature channels in convolutional layers is set to 180. A dual-branch shared progressive super-resolution strategy is adopted: for a scaling factor of 4, two $\times 2$ super-resolution stages are performed; for a scaling factor of 8, $\times 2$ super-resolution is performed first, followed by $\times 4$ super-resolution. The upsampling layers adopt pixel shuffle (subpixel convolution) layers. The training batch sizes for scaling factors of 4 and 8 are set to 32 and 12, respectively. The optimizer used is Adam (initial learning rate of 2×10^{-4}, decayed by a factor of 10 after 20 epochs). The total number of training epochs is 30. All methods are implemented based on PyTorch and trained/tested on a computer equipped with an Nvidia RTX 3060Ti GPU.

This article adopts six widely used super-resolution evaluation metrics: peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), correlation coefficient (CC), root-mean-square error (RMSE), structural similarity index (SSIM), and erreur relative globale adimensionnellede synthese (ERGAS). The optimal values for these metrics are: $+\infty$ for PSNR, 0 for SAM, 1 for CC, 0 for RMSE, 1 for SSIM, and 0 for ERGAS.

B. Comparison of Different Algorithms

1) Comparison on Chikusei Dataset: After removing the edges, the effective central region size of the Chikusei dataset is $2304 \times 2048 \times 128$. Following the processing method in [34] and [35], this article crops four nonoverlapping subimages of size $512 \times 512 \times 128$ from the top region as the test set. The remaining area is used as the training and validation sets (with

[1] https://hyperspectral.ee.uh.edu/?page_id=1075

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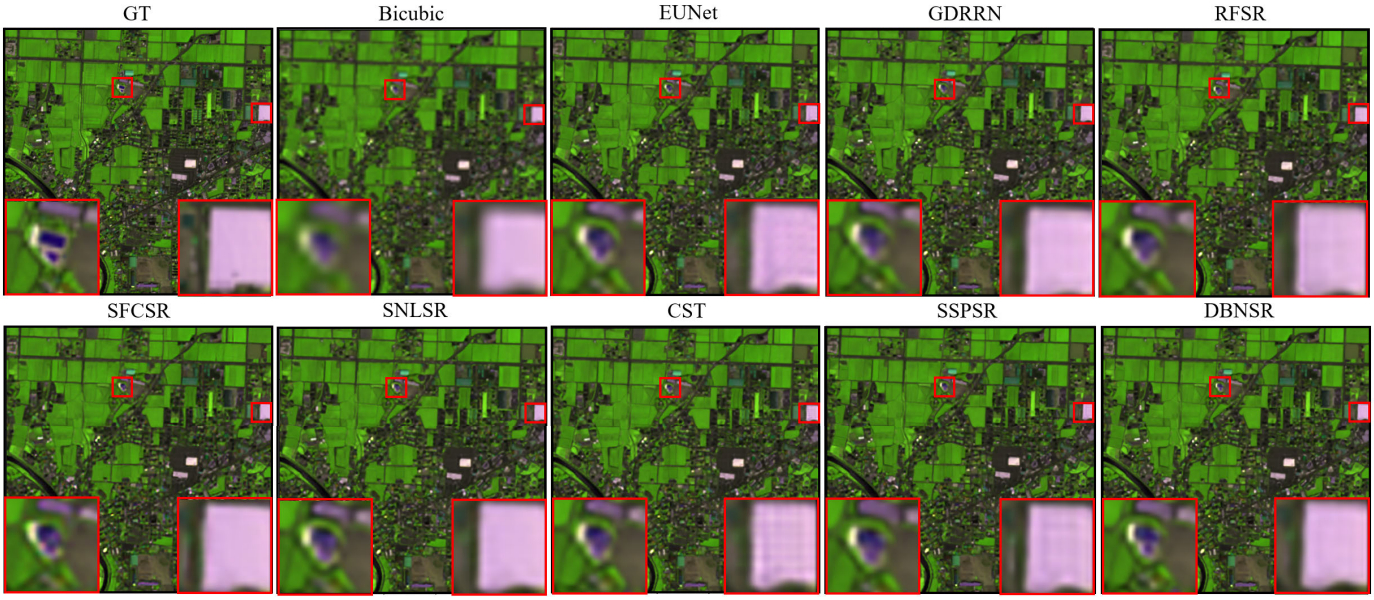


Fig. 7. Reconstructed high-resolution HSI results of different methods on part of the Chikusei test set and the corresponding GT. The images are visualized using spectral bands 70, 100, and 36 as R–G–B, with a super-resolution scaling factor of ×4.

TABLE I

QUANTITATIVE PERFORMANCE ON THE CHIKUSEI DATASET. BOLD INDICATES THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS. *d* REFERS TO THE SCALE FACTOR

Methods	*d*	CC↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SSIM↑
Bicubic	×4	0.9212	3.4039	0.0156	6.7563	37.6377	0.8953
RFSR	×4	0.9512	2.5035	0.0121	5.2451	39.8476	0.9373
SFCSR	×4	0.9404	3.2584	0.0136	5.7796	38.8945	0.9221
GDRRN	×4	0.9477	2.7370	0.0125	5.4763	39.4999	0.9326
EUNet	×4	0.9502	2.4647	0.0122	5.3083	39.7464	0.9363
SSPSR	×4	<u>0.9562</u>	2.3863	<u>0.0115</u>	<u>4.9497</u>	<u>40.3440</u>	<u>0.9433</u>
SNLSR	×4	0.9495	2.4836	0.0123	5.3493	39.6787	0.9347
CST	×4	0.9543	<u>2.3859</u>	0.0117	5.0822	40.1285	0.9415
DBNSR	×4	**0.9568**	**2.3408**	**0.0113**	**4.9326**	**40.3806**	**0.9442**
Bicubic	×8	0.8313	5.0435	0.0224	4.8487	34.5048	0.8068
RFSR	×8	0.8614	4.4025	0.0202	4.3798	35.3639	0.8351
SFCSR	×8	0.8537	4.9957	0.0212	4.4807	35.0735	0.8250
GDRRN	×8	0.8658	4.4149	0.0199	4.3246	35.4794	0.8395
EUNet	×8	0.8680	4.0474	0.0197	4.2909	35.5948	0.8460
SSPSR	×8	<u>0.8772</u>	<u>3.9752</u>	<u>0.0193</u>	<u>4.1279</u>	<u>35.8341</u>	<u>0.8534</u>
SNLSR	×8	0.8683	4.1868	0.0197	4.2971	35.5552	0.8436
CST	×8	0.8725	4.0663	0.0194	4.2467	35.6718	0.8495
DBNSR	×8	**0.8787**	**3.9434**	**0.0190**	**4.1138**	**35.8926**	**0.8564**

10% of the training data randomly selected as the validation set). For ×4 super-resolution, each GT sample in the training set is obtained by a sliding window of size 64 × 64 with a stride of 32, and the corresponding low-resolution HSI of size 16 × 16 is generated by bicubic downsampling. For ×8 super-resolution, each GT sample in the training set is obtained by a sliding window of size 128 × 128 with a stride of 64, and the corresponding low-resolution HSI of size 16 × 16 is generated by bicubic downsampling.

Table I shows the quantitative results of different methods on the Chikusei dataset under various scaling factors. The best results are shown in bold, and the second-best results are underlined. It can be seen that learning-based super-resolution methods achieve significant improvements compared with the basic bicubic interpolation upsampling method. The abundance-map-based super-resolution and original HSI-based super-resolution methods achieve super-resolution reconstruction by exploring spectral and spatial information, respectively, but neither effectively extracts the spatial–spectral information of HSI simultaneously. Notably, by mining spatial–spectral information through a dual-branch architecture, the proposed DBNSR achieves the best performance on all metrics at scaling factors ×4 and ×8, demonstrating the effectiveness and superiority of DBNSR.

Fig. 7 presents qualitative results of all methods on the Chikusei dataset with a scaling factor of ×4. The displayed images are visualized using bands 70, 100, and 36 as the R–G–B channels. It can be seen that the bicubic upsampling super-resolution results are relatively blurry, with overall details overly smoothed. Recent methods, such as SSPSR and CST, have made progress in restoring the overall structure but still suffer from unclear boundaries and severe artifacts. In contrast, the proposed DBNSR reconstructs high-resolution image details more clearly and sharply compared to other methods.

Moreover, zoomed-in views of representative high-frequency and low-frequency spatial detail regions in the red boxes are shown, with their corresponding mean spectral error curves presented in Fig. 8. It is evident that DBNSR outperforms other methods in super-resolution quality at high-frequency regions while maintaining spectral consistency in low-frequency areas, effectively reducing artifact generation.

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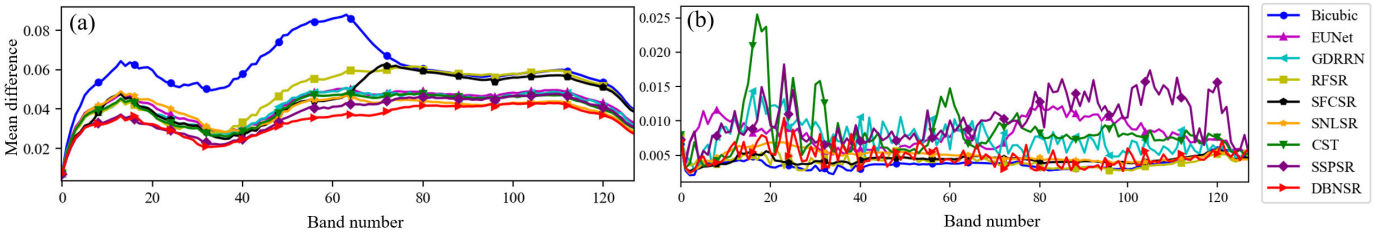


Fig. 8. Average spectral difference curves of different methods in the red boxed areas of Fig. 5. (a) Corresponds to the left red box (high-frequency detail region). (b) Corresponds to the right red box (low-frequency region).

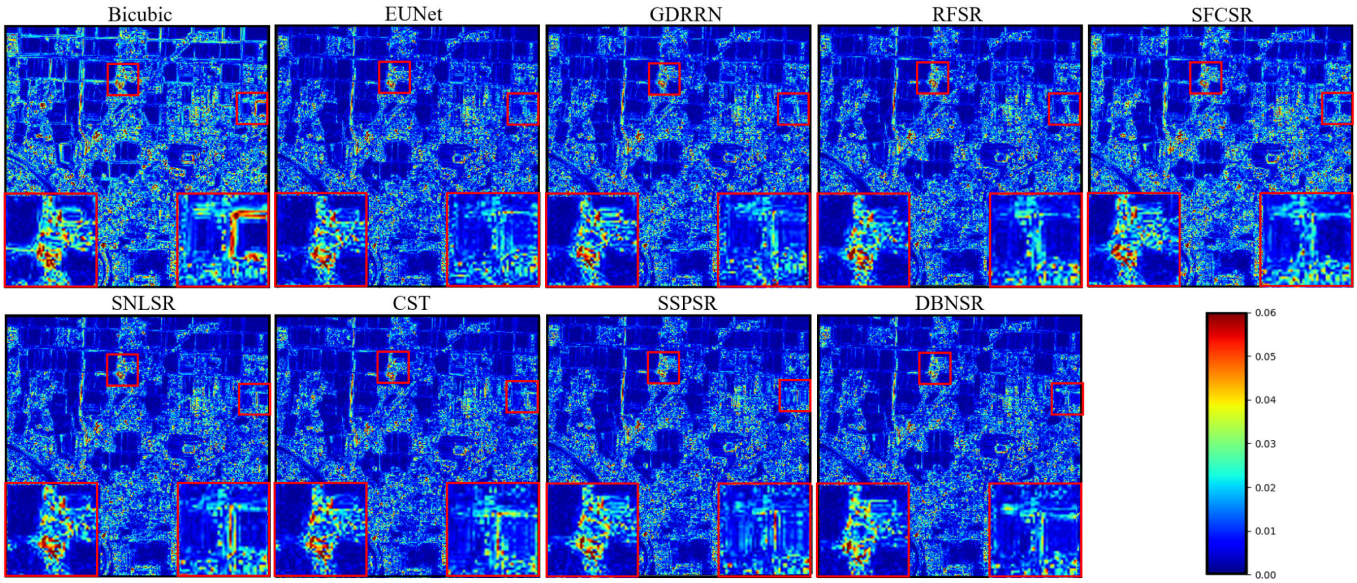


Fig. 9. Error maps of different methods on part of the Chikusei test set, with a super-resolution scaling factor of $\times 4$.

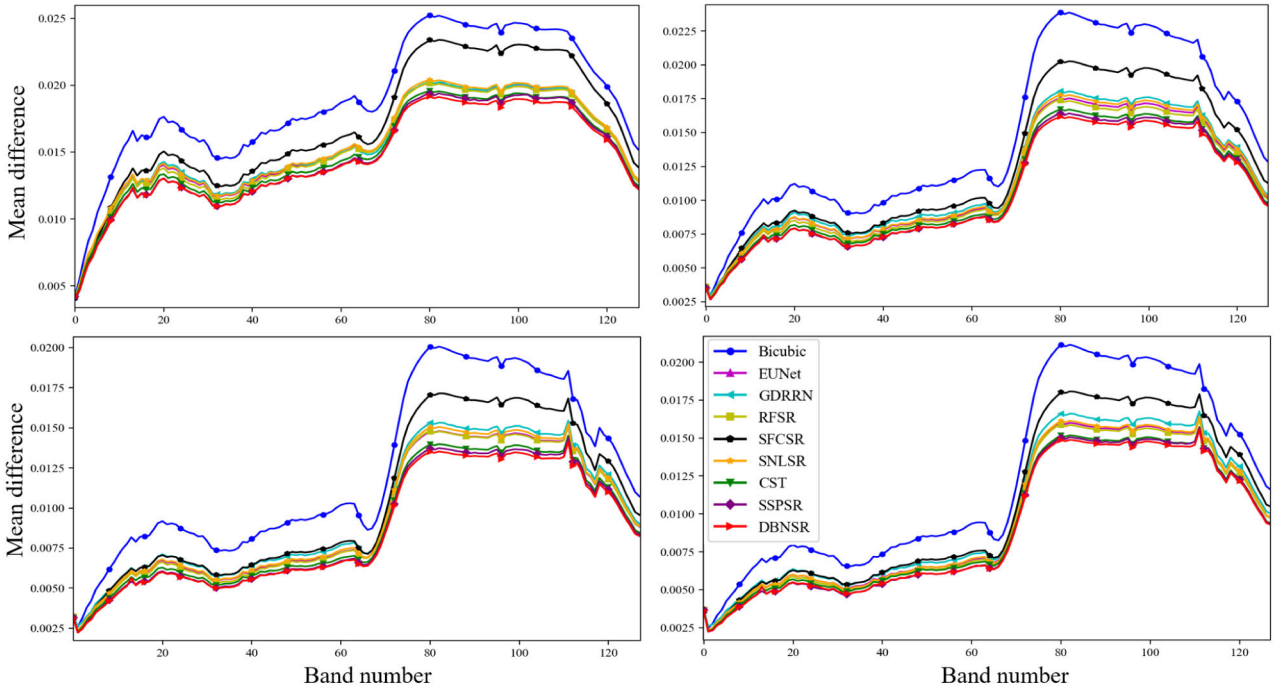


Fig. 10. Mean spectral difference curves of different methods on four Chikusei test samples with a super-resolution scaling factor of $\times 4$.

In addition, Fig. 9 visualizes the mean spectral error images for the entire scene, providing an intuitive pixel-wise reconstruction error map. In the error images, blue indicate higher reconstruction accuracy. The proposed method exhibits the smallest errors, indicating its superiority.

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TABLE II
QUANTITATIVE PERFORMANCE ON THE HOUSTON DATASET. BOLD INDICATES THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS. d REFERS TO THE SCALE FACTOR

Methods	d	CC↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SSIM↑
Bicubic	×4	0.9952	1.1087	0.0056	1.1699	46.4698	0.9840
RFSR	×4	0.9976	0.8321	0.0039	0.8176	49.1838	0.9918
SFCSR	×4	0.9976	0.8577	0.0039	0.8387	48.9804	0.9915
GDRRN	×4	0.9972	0.9942	0.0043	0.9017	48.2751	0.9903
EUNet	×4	0.9976	0.8352	0.0039	0.8199	49.4085	0.9917
SSPSR	×4	0.9978	0.8472	0.0038	0.7928	49.4044	0.9925
SNLSR	×4	0.9976	0.8348	0.0039	0.8198	49.3059	0.9918
CST	×4	0.9978	0.8393	0.0038	0.7943	49.3461	0.9924
DBNSR	×4	**0.9979**	**0.8290**	**0.0037**	**0.7693**	**49.4979**	**0.9929**
Bicubic	×8	0.9746	2.6948	0.0132	1.3590	39.2298	0.9323
RFSR	×8	0.9851	2.0626	0.0099	1.0307	41.1781	0.9546
SFCSR	×8	0.9837	2.2838	0.0104	1.0792	40.8027	0.9504
GDRRN	×8	0.9811	2.5013	0.0112	1.1701	39.9929	0.9452
EUNet	×8	0.9847	2.0643	0.0101	1.0448	41.4071	0.9529
SSPSR	×8	0.9855	2.0184	0.0099	1.0112	41.3369	0.9562
SNLSR	×8	0.9846	2.0861	0.0101	1.0473	41.3256	0.9529
CST	×8	0.9856	2.0367	0.0098	1.0109	41.4168	0.9563
DBNSR	×8	**0.9859**	**1.9767**	**0.0096**	**1.0001**	**41.4724**	**0.9568**

TABLE III
QUANTITATIVE PERFORMANCE ON THE PAVIA DATASET. BOLD INDICATES THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS. d REFERS TO THE SCALE FACTOR

Methods	d	CC↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SSIM↑
Bicubic	×4	0.8572	6.1217	0.0438	6.9181	27.5627	0.7209
RFSR	×4	0.8950	5.5109	0.0367	5.9193	28.9817	0.8075
SFCSR	×4	0.8777	6.6842	0.0403	6.3636	28.2875	0.7806
GDRRN	×4	0.8865	6.3481	0.0382	6.1525	28.6444	0.7933
EUNet	×4	0.8899	5.5553	0.0377	6.0594	28.7684	0.7997
SSPSR	×4	0.8956	5.6576	0.0366	5.8755	29.0278	0.8109
SNLSR	×4	0.8845	5.9633	0.0388	6.1876	28.5601	0.7879
CST	×4	0.8940	5.6806	0.0370	5.9484	28.9269	0.8057
DBNSR	×4	**0.8998**	**5.4001**	**0.0359**	**5.7841**	**29.1745**	**0.8173**
Bicubic	×8	0.6955	7.8094	0.0626	4.8165	24.6405	0.4618
RFSR	×8	0.7229	7.5552	0.0594	4.5833	25.0701	0.5107
SFCSR	×8	0.7103	7.6824	0.0608	4.6780	24.8905	0.5053
GDRRN	×8	0.7157	7.9533	0.0598	4.6371	24.9806	0.5117
EUNet	×8	0.7259	7.4495	0.0589	4.5635	25.1171	0.5228
SSPSR	×8	0.7328	7.5825	0.0584	4.5292	25.1827	0.5356
SNLSR	×8	0.7165	7.6985	0.0601	4.6289	24.9835	0.5042
CST	×8	0.7262	7.6054	0.0588	4.5644	25.1182	0.5246
DBNSR	×8	**0.7398**	**7.2650**	**0.0576**	**4.4715**	**25.2975**	**0.5488**

Finally, Fig. 10 shows the mean spectral error curves for four nonoverlapping test images at a scaling factor of ×4 for different methods, offering spectral-perspective analysis of the super-resolution results. Lower curves indicate less spectral difference and distortion between the super-resolution results and GT. Clearly, the proposed method achieves the best spectral reconstruction results across different test images.

2) Comparison on Houston Dataset: For the Houston dataset, this article extracts the central region of size $1024 \times 4096 \times 48$ as the training and testing data. Among these, four nonoverlapping subimages of size $512 \times 512 \times 48$ are cropped from the left side of the dataset as the test set. The remaining region is processed similar to the Chikusei dataset to obtain training and validation datasets for scaling factors ×4 and ×8 (with 10% of the training data randomly selected as the validation set).

Table II presents the quantitative results of all methods on the Houston dataset under different scaling factors. It can be observed that, due to the Houston dataset providing more training data and fewer spectral bands, the overall metrics of all methods outperform those on other datasets. Meanwhile, for all evaluation metrics at different scaling factors, the proposed DBNSR method surpasses other methods. This indicates that the dual-branch combined network has strong adaptability, further validating the effectiveness of DBNSR and highlighting the superiority of the spatial–spectral information extraction approach that combines abundance maps and original HSI in hyperspectral image super-resolution reconstruction.

Similarly, as shown in Fig. 11, qualitative results of all methods on part of the Houston test data with a scaling factor of ×4 are presented. The displayed images use spectral bands 29, 26, and 19 as the R–G–B channels for visualization. It can be seen that other methods inevitably produce blurry edges, while the proposed method better restores both low-frequency and high-frequency information in the super-resolved HSI. Furthermore, the zoomed-in views within the red boxes further emphasize the advantages of the proposed method.

Finally, as shown in Fig. 12, mean spectral error curves for four nonoverlapping Houston test images at a scaling factor of ×4 are also visualized for different methods. Clearly, the proposed method achieves the best spectral reconstruction results across different test images.

3) Comparison on Pavia Dataset: For the Pavia dataset, this article extracts the central region of size $1088 \times 704 \times 102$ as the training and testing data. Among these, five nonoverlapping subimages of size $192 \times 192 \times 102$ are cropped from the left side of the dataset as the test set. The remaining area is processed similar to the Chikusei dataset to obtain training and validation datasets for scaling factors ×4 and ×8 (with 10% of the training data randomly selected as the validation set).

Table III presents the quantitative results of all methods on the Pavia dataset under different scaling factors. Compared with other datasets, Pavia has lower spatial resolution and limited training samples, resulting in relatively poorer performance of all methods on this dataset. Nevertheless, the proposed DBNSR method still outperforms other methods on all metrics. This indicates that even with limited samples, the

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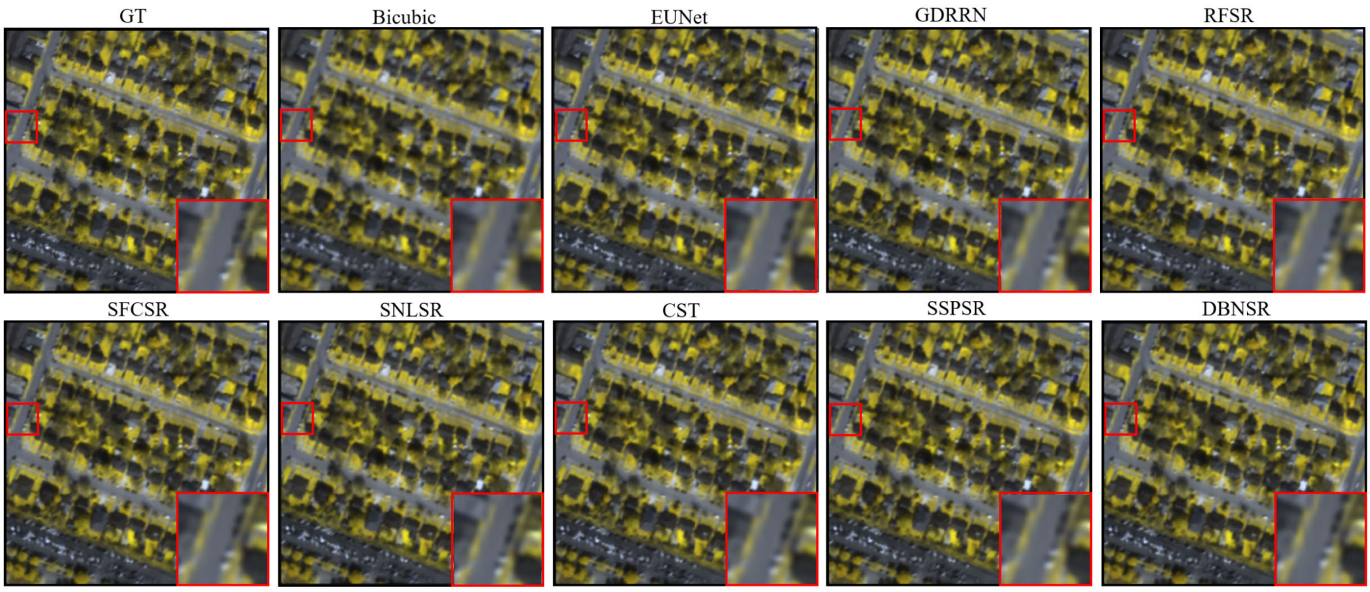


Fig. 11. Reconstructed high-resolution HSI results of different methods on part of the Houston test set and the corresponding GT. The images are visualized using spectral bands 29, 26, and 19 as R–G–B, with a super-resolution scaling factor of $\times 4$.

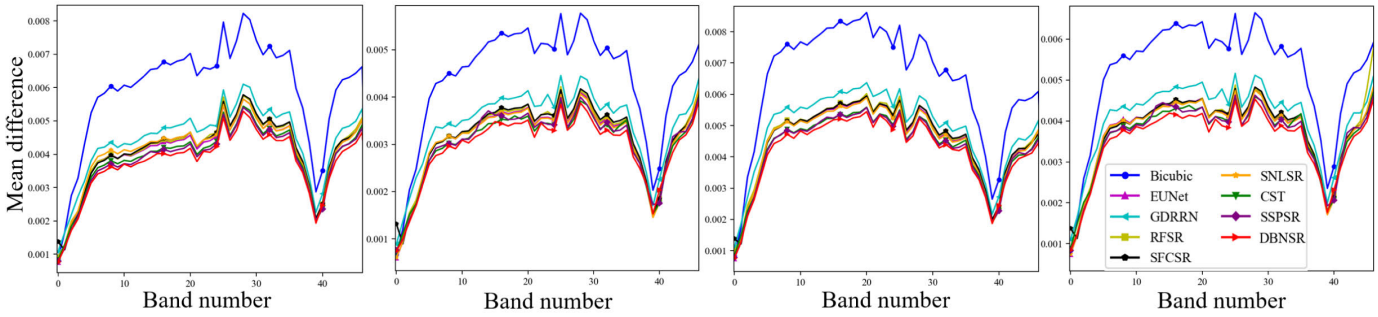


Fig. 12. Mean spectral difference curves of different methods on four Houston test samples with a super-resolution scaling factor of $\times 4$.

proposed method effectively mines spatial–spectral information through the dual-branch spatial–spectral extraction and dedicated high-frequency detail extraction strategy, maintaining competitive advantages on this dataset.

As shown in Fig. 13, qualitative results of all methods on part of the Pavia test data with a scaling factor of $\times 4$ are presented. The displayed images use spectral bands 100, 30, and 12 as the R–G–B channels for visualization. It can be seen that due to limited training samples, many methods exhibit blurry edges and artifacts, while DBNSR better restores both low-frequency and high-frequency information in the super-resolved HSI. Furthermore, the zoomed-in views within the green boxes further highlight the advantages of the proposed method.

As shown in Fig. 14, mean spectral error curves for five nonoverlapping Pavia test images at a scaling factor of $\times 4$ are visualized for different methods. Clearly, the proposed method also achieves the best spectral reconstruction results across different test images.

4) Model Complexity Analysis: This article compares the model complexity of different methods in terms of the number of parameters, FLOPs, and inference time, with the tested input size set to $128 \times 16 \times 16$ and a scaling factor of $\times 4$. As shown in Table IV, compared to faster super-resolution reconstruction methods such as GDRRN, EUNet, and RFSR, DBNSR exhibits slightly slower inference speed but demonstrates significant advantages in performance metrics. Although SNLSR achieves lower complexity and a smaller model size, its inference speed is affected by capturing nonlocal spectral relationships. CST improves model accuracy significantly by capturing nonlocal spatial–spectral information but also impacts inference speed. SFCSR reduces the number of parameters through a grouping strategy, but the large number of groups leads to slower overall inference speed. SSPSR enhances super-resolution accuracy at the cost of a larger model and increased complexity. The proposed DBNSR improves the mining of spatial–spectral information by utilizing a dual-branch network based on abundance maps and HSI, enhancing the super-resolution accuracy. At the same time, it achieves efficient spatial–spectral information extraction through a parameter-sharing strategy in the dual-branch network. Overall, the proposed DBNSR method maintains a reasonable number of parameters and relatively low FLOPs while achieving faster inference speed. This indicates that it achieves a better balance between efficiency and performance.

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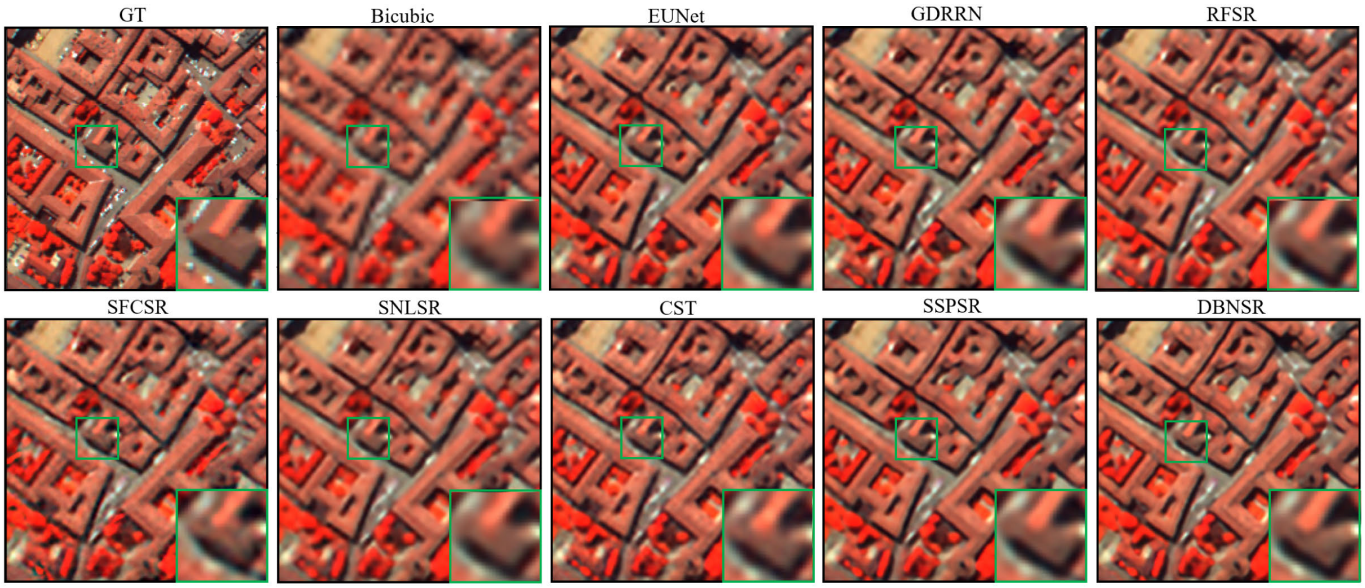


Fig. 13. Reconstructed high-resolution HSI results of different methods on part of the Pavia test set and the corresponding GT. The images are visualized using spectral bands 100, 30, and 12 as R–G–B, with a super-resolution scaling factor of ×4.

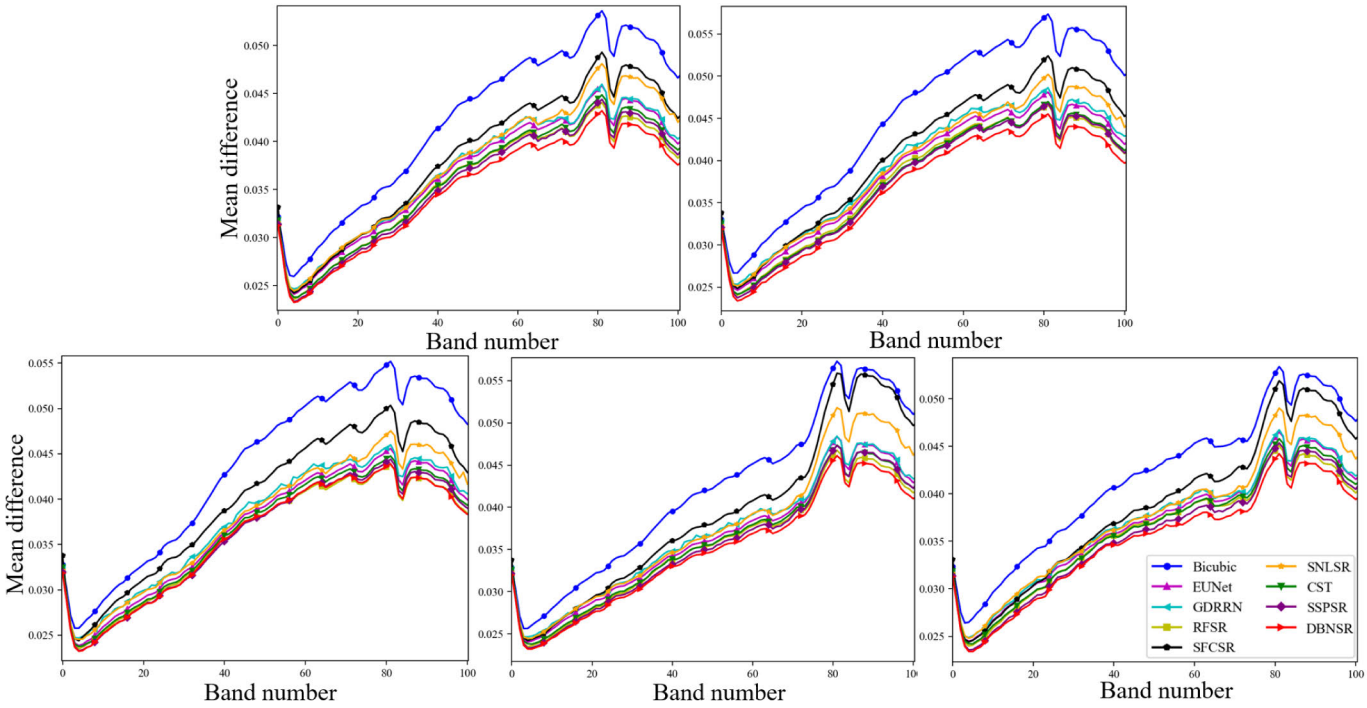


Fig. 14. Mean spectral difference curves of different methods on five Pavia test samples with a super-resolution scaling factor of ×4.

C. Ablation Study

In this section, ablation studies on SSRB and SSF are conducted across different datasets and scaling factors. The remaining ablation studies are performed on the Chikusei dataset with a scaling factor of ×4, using various model variants to validate the effectiveness of the proposed DBNSR method.

1) Effectiveness of the HFE Module: In the DBNSR model, a dedicated HFE module is designed, which contains multiple cascaded SSRBs as the basic components for feature extraction.

TABLE IV

COMPARISON OF MODEL PARAMETERS, FLOPS, AND INFERENCE TIME FOR DIFFERENT METHODS

Methods	EUNet	GDRNN	RFSR	SFCSR	SNLSR	CST	SSPSR	DBNSR
Params(M)	0.84	0.44	0.98	1.24	1.65	11.1	13.6	11.1
FLOPs(G)	4.33	0.42	3.78	90.7	0.86	5.32	43.5	35.8
Time(ms)	177	105	169	1050	205	446	193	190

As shown in Fig. 15, to verify the effectiveness of the proposed SSRB, several DBNSR variants were designed by

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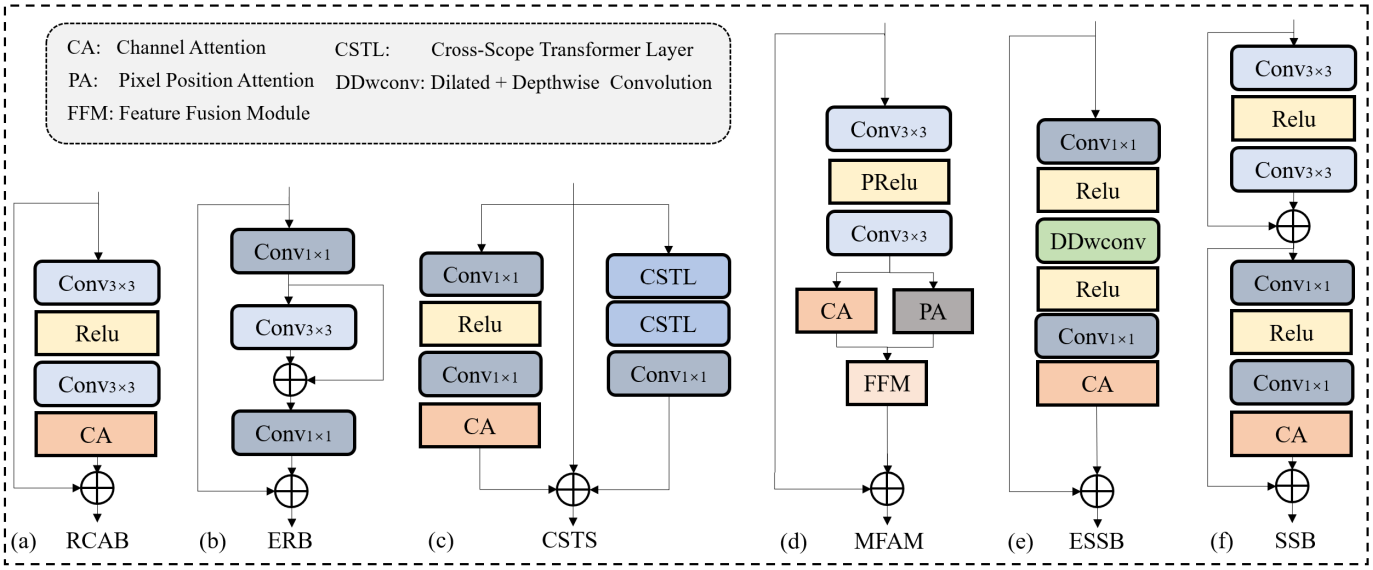


Fig. 15. Comparison of basic modules in different super-resolution networks. (a) RCAN, (b) FMEN, (c) CST, (d) CTE-Net, (e) EUNet, and (f) SSPSR.

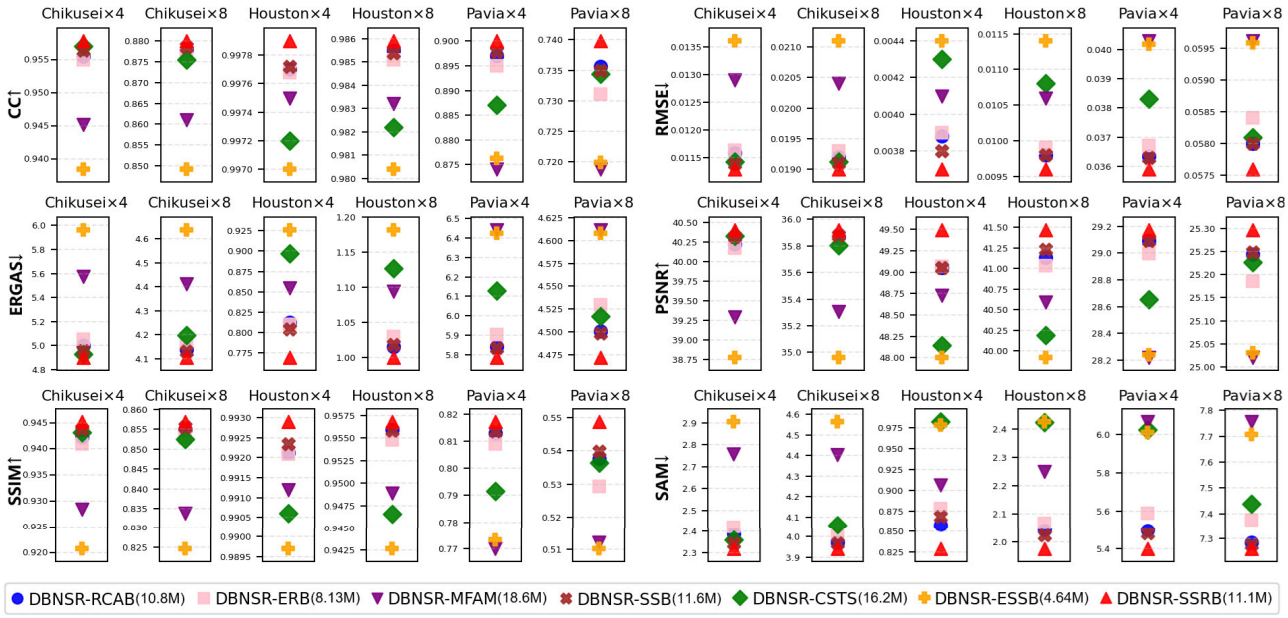


Fig. 16. Ablation study. Quantitative results of DBNSR variants with different spatial–spectral feature extraction modules under different scale factors on various test sets. The numbers in parentheses indicate the parameter size of each variant for an input of 128 × 16 × 16 at a ×4 scale factor.

replacing the basic modules with those from different methods for comparison. Specifically, these variants are given as follows.

1) *DBNSR-RCAB:* SSRB is replaced with RCAB, the basic feature extraction module in RCAN [21].
2) *DBNSR-ERB:* SSRB is replaced with ERB, the basic feature extraction module in FMEN [50].
3) *DBNSR-CSTS:* SSRB is replaced with CSTS, the basic feature extraction module in CST [35].
4) *DBNSR-MFAM:* SSRB is replaced with MFAM, the basic feature extraction module in CETNet [49].
5) *DBNSR-ESSB:* SSRB is replaced with ESSB, the basic feature extraction module in EUNet [28].
6) *DBNSR-SSB:* SSRB is replaced with SSB, the basic feature extraction module in SSPSR [34].

As shown in Fig. 16, the proposed SSRB module achieves the best performance across different datasets and scaling factors compared with other variants, demonstrating the effectiveness of the proposed HFE structure. Meanwhile, the comparison of parameter numbers among different variants shows that the SSRB module achieves a better balance between performance and parameter size, reflecting its efficiency.

2) Effectiveness of the SSF Module: In the proposed DBNSR model, the spatial–spectral feature fusion (SSF) module is a key component that effectively integrates the results from the dual-branch network. As shown in Fig. 17,

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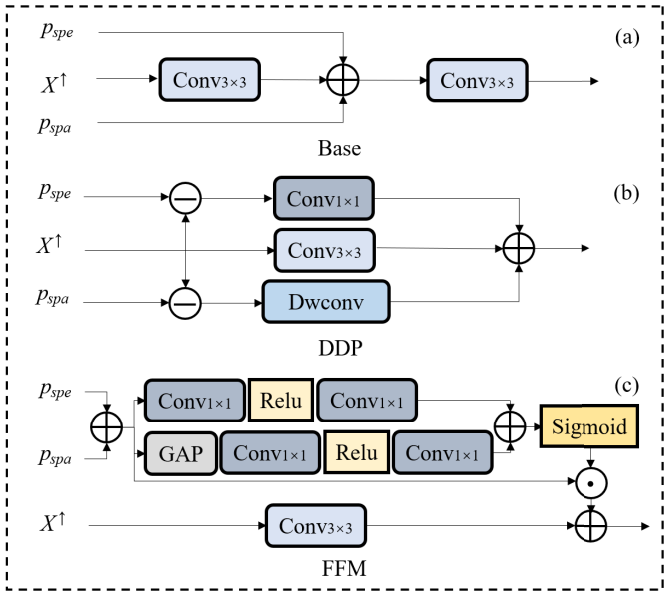


Fig. 17. Comparison of fusion modules in different super-resolution networks. (a) Base, (b) SNLSR, and (c) CET-Net.

to verify the effectiveness of SSF, several DBNSR variants were designed based on current spatial–spectral feature fusion strategies for comparison. These variants are given as follows.

1) *DBNSR-Base:* SSF is replaced with Base, a common feature fusion module used in typical super-resolution networks [34].
2) *DBNSR-DDP:* SSF is replaced with DDP, the SSF module used in SNLSR [37].
3) *DBNSR-FFM:* SSF is replaced with FFM, the SSF module used in CETNet [49].

Fig. 18 presents the results under different fusion strategies. It can be seen that the proposed SSF module achieves the best super-resolution performance across all datasets and scaling factors, indicating its superior fusion capability. In contrast, the commonly used Base fusion strategy leads to degraded super-resolution results. The DDP and FFM fusion modules fail to consider the complementary information between spatial and spectral features, which is detrimental to spectral continuity. By comparison, the SSF module, which is built upon the output of the abundance-map super-resolution network, better preserves spectral continuity after super-resolution and is more suitable for complementary spatial–spectral features fusion. In addition, the comparison of parameter sizes across different variants further verifies the effectiveness and efficiency of the proposed SSF module.

3) Effectiveness of the Dual-Branch Network Structure and Parameter Sharing: To study the impact of the DBNSR dual-branch network and parameter sharing strategy, corresponding ablation experiments were conducted. Accordingly, several variants were introduced.

1) *DBNSR w/o Org (Original-Image Branch):* The original HSI branch, one of the dual branches, is removed.
2) *DBNSR w/o Abd (Abundance-Map Branch):* Similarly, the abundance-map branch, one of the dual branches, is removed.

TABLE V

ABLATION STUDY. QUANTITATIVE RESULTS OF DIFFERENT DBNSR VARIANTS ON THE CHIKUSEI TEST SET AT SCALE FACTOR ×4

Variants	Params	CC↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SSIM↑
DBNSR w/o org	10.8M	0.9512	2.5184	0.0121	5.2726	39.8256	0.9379
DBNSR w/o abd	10.9M	0.9562	2.3531	0.0116	4.9489	40.3392	0.9436
DBNSR-A2O	11.5M	0.9508	2.5146	0.0122	5.2604	39.8052	0.9371
DBNSR-O2A	1.4M	0.9528	2.4540	0.0119	5.1811	39.9832	0.9395
DBNSR w/o sha	21.2M	0.9566	2.3411	**0.0113**	4.9332	40.3804	0.9441
DBNSR	11.1M	**0.9568**	**2.3408**	**0.0113**	**4.9326**	**40.3806**	**0.9442**

TABLE VI

ABLATION STUDY. QUANTITATIVE RESULTS OF DIFFERENT NUMBERS OF ENDMEMBERS ON THE CHIKUSEI TEST SET AT SCALE FACTOR ×4

Variants	Params	CC↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SSIM↑
DBNSR-C5	10.4M	0.9536	2.4007	0.0119	4.9916	40.2533	0.9411
DBNSR-C10	10.6M	0.9548	2.3945	0.0116	4.9765	40.2896	0.9426
DBNSR-C20	10.8M	0.9563	2.3638	0.0114	4.9644	40.3060	0.9434
DBNSR-C30	11.1M	**0.9568**	**2.3408**	**0.0113**	4.9326	40.3806	**0.9442**
DBNSR-C40	11.5M	**0.9568**	2.3449	**0.0113**	**4.9325**	**40.3813**	0.9438

3) *DBNSR-A2O:* First perform ×2 super-resolution on the abundance map, followed by ×2 super-resolution on the original image.
4) *DBNSR-O2A:* First perform ×2 super-resolution on the original image, followed by ×2 super-resolution on the abundance map.
5) *DBNSR w/o SHA (Share):* The parameter sharing strategy in DBNSR is removed in this variant.

The experimental results shown in Table V indicate that removing either branch causes a decline in overall network performance, demonstrating the effectiveness of the dual-branch strategy. Furthermore, sequentially executing the two branches results in a noticeable performance drop, indicating the effectiveness and rationality of the parallel dual-branch structure, which better leverages the complementary advantages of the original-image branch and the abundance-map branch for super-resolution. In addition, when parameter sharing is removed, the number of network parameters nearly doubles, while the model performance exhibits slight but consistently downward fluctuations in some metrics. This indicates that parameter sharing facilitates the synchronous learning of HSI and abundance-map representations in the dual-branch network, thereby improving model efficiency without sacrificing accuracy.

4) Effectiveness of the Number of Endmembers: The number of endmembers is a critical factor in the design of the unmixing module, and selecting an appropriate value is essential for improving the overall performance of the model. Table VI presents the unmixing results under different numbers of endmembers. As observed, increasing the number of endmembers consistently enhances the model's performance across all evaluation metrics, but it also leads to an increase in model parameters. When the number of endmembers exceeds

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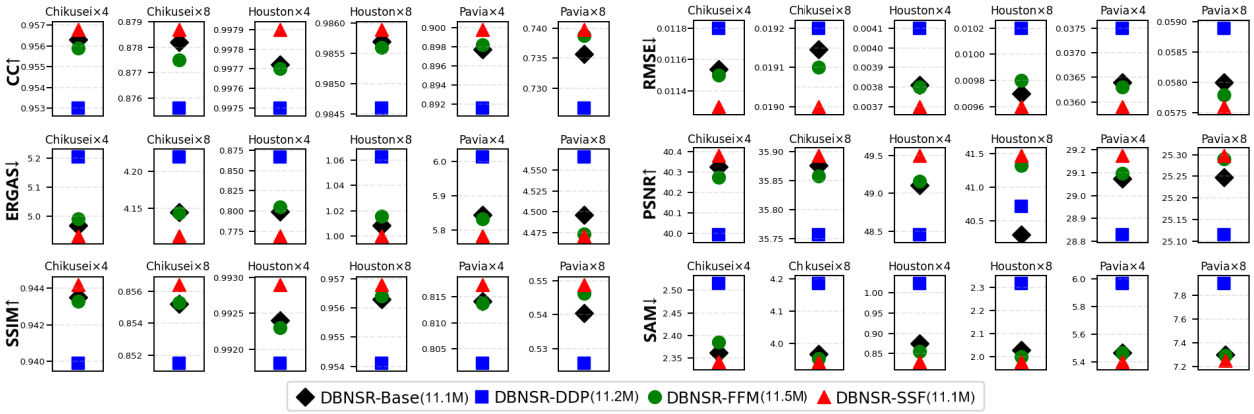


Fig. 18. Ablation study. Quantitative results of DBNSR variants with different spatial–spectral feature fusion modules under different scale factors on various test sets. The numbers in parentheses indicate the parameter size of each variant for an input of 128 × 16 × 16 at a ×4 scale factor.

TABLE VII
QUANTITATIVE PERFORMANCE ON THE CHIKUSEI DATASET. BOLD INDICATES THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS. *d* REFERS TO THE SCALE FACTOR

Methods	*d*	CC↑	SAM↓	RMSE↓	ERGAS↓	PSNR↑	SSIM↑
Bicubic	×16	0.7193	6.5909	0.0282	3.0285	32.5373	0.7587
RFSR	×16	0.7390	6.3726	0.0273	2.9120	32.8310	0.7634
SFCSR	×16	0.7325	6.7166	0.0281	2.9640	32.6507	0.7595
GDRRN	×16	0.7328	6.4142	0.0274	2.9884	32.7173	0.7637
EUNet	×16	0.7422	5.8446	0.0272	2.9448	32.8207	0.7723
SSPSR	×16	<u>0.7625</u>	<u>5.7399</u>	<u>0.0263</u>	<u>2.8475</u>	<u>33.1106</u>	<u>0.7796</u>
SNLSR	×16	0.7336	6.1110	0.0273	2.9881	32.7293	0.7669
CST	×16	0.7529	5.8738	0.0266	2.8579	33.0217	0.7732
DBNSR	×16	**0.7631**	**5.6822**	**0.0261**	**2.8185**	**33.1726**	**0.7819**

30, the performance gains become marginal, while the parameter growth becomes more noticeable. Therefore, considering the tradeoff between performance and complexity, choosing 30 endmembers is a reasonable configuration.

V. DISCUSSION

In this section, we provide a further analysis of the model's performance under extreme scaling factors. As shown in Table VII, even under this highly challenging setting, the proposed DBNSR maintains a relatively stable competitive advantage, demonstrating strong robustness in both spatial and spectral reconstruction. However, due to the substantial loss of high-frequency information, the absolute performance of all methods exhibits an expected decline. This observation highlights the inherent limitations of "single-image super-resolution" in extreme magnification scenarios, where severe information deficiency inevitably constrains reconstruction capability. Therefore, developing specialized frameworks tailored for ultrahigh magnification super-resolution is of great importance. Among potential solutions, incorporating multimodal guidance—such as high-resolution RGB images—offers a feasible pathway for compensating or generating missing frequency-domain information, and represents a promising direction for future breakthroughs.

VI. CONCLUSION

This article investigates single-image spatial super-resolution for HSIs. First, a DBNSR based on abundance maps and the original HSI is proposed. The abundance-map-based super-resolution branch obtains abundance maps and endmembers through unmixing, performs super-resolution on the abundance maps to generate high-resolution abundance maps and then reconstructs the high-resolution HSI by linear mixing of the high-resolution abundance maps and endmembers. The original HSI-based super-resolution branch directly performs spatial super-resolution on the HSI to obtain a high-resolution HSI. Second, an SSF module is proposed to fuse the results from the dual branches. Then, an HFE module is introduced to extract high-frequency information from both the abundance maps and HSI. By leveraging the dual-branch network and spatial–spectral feature extraction modules, effective spatial–spectral information extraction of HSI is achieved, enhancing the model's super-resolution performance. Furthermore, the parameter sharing strategy in the dual-branch network improves super-resolution performance and enhances the model's efficiency. Finally, the proposed DBNSR is compared with current popular methods. Experimental results validate the superiority of the proposed method, and ablation studies verify the effectiveness of the proposed improvements.

REFERENCES

[1] L. Fang, Y. Jiang, Y. Yan, J. Yue, and Y. Deng, "Hyperspectral image instance segmentation using spectral–spatial feature pyramid network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502613.

[2] Y. Xu, L. Zhang, B. Du, and L. Zhang, "Hyperspectral anomaly detection based on machine learning: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3351–3364, 2022.

[3] Y. Tan, L. Lu, L. Bruzzone, R. Guan, Z. Chang, and C. Yang, "Hyperspectral band selection for lithologic discrimination and geological mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 471–486, 2020.

[4] Z. Wang, J. Chen, and S. C. H. Hoi, "Deep learning for image super-resolution: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3365–3387, Oct. 2021.

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[5] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[6] Q. Qu, B. Pan, X. Xu, T. Li, and Z. Shi, "Unmixing guided unsupervised network for RGB spectral super-resolution," *IEEE Trans. Image Process.*, vol. 32, pp. 4856–4867, 2023.

[7] Z. Guo, J. Xin, N. Wang, J. Li, and X. Gao, "External-internal attention for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538714.

[8] Z. Guo, J. Xin, N. Wang, J. Li, X. Wang, and X. Gao, "Unsupervised across domain consistency- difference network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511613.

[9] J. Li et al., "Deep learning in multimodal remote sensing data fusion: A comprehensive review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102926.

[10] W. Dong, J. Qu, S. Xiao, T. Zhang, Y. Li, and X. Jia, "Noise prior knowledge informed Bayesian inference network for hyperspectral super-resolution," *IEEE Trans. Image Process.*, vol. 32, pp. 3121–3135, 2023.

[11] K. Zheng, L. Gao, D. Hong, B. Zhang, and J. Chanussot, "NonRegSRNet: A nonrigid registration hyperspectral super-resolution network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520216.

[12] H. Huang, J. Yu, and W. Sun, "Super-resolution mapping via multi-dictionary based sparse representation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2014, pp. 3523–3527.

[13] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3862–3871.

[14] L. Zhang, W. Wei, Y. Zhang, C. Tian, and F. Li, "Reweighted Laplace prior based hyperspectral compressive sensing for unknown sparsity," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 2274–2281.

[15] T. Akgun, Y. Altunbasak, and R. M. Mersereau, "Super-resolution reconstruction of hyperspectral images," *IEEE Trans. Image Process.*, vol. 14, no. 11, pp. 1860–1875, Nov. 2005.

[16] S. He, H. Zhou, Y. Wang, W. Cao, and Z. Han, "Super-resolution reconstruction of hyperspectral images via low rank tensor modeling and total variation regularization," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 6962–6965.

[17] Y. Wang, X. Chen, Z. Han, and S. He, "Hyperspectral image super-resolution via nonlocal low-rank tensor approximation and total variation regularization," *Remote Sens.*, vol. 9, no. 12, p. 1286, Dec. 2017.

[18] H. Irmak, G. B. Akar, and S. E. Yuksel, "A MAP-based approach for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2942–2951, Jun. 2018.

[19] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1132–1140.

[20] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

[21] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image superresolution using very deep residual channel attention networks," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 294–310.

[22] Y. Yuan, X. Zheng, and X. Lu, "Hyperspectral image superresolution by transfer learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 1963–1974, May 2017.

[23] S. Mei, X. Yuan, J. Ji, Y. Zhang, S. Wan, and Q. Du, "Hyperspectral image spatial super-resolution via 3D full convolutional neural network," *Remote Sens.*, vol. 9, no. 11, p. 1139, Nov. 2017.

[24] Z. Liu, W. Wang, Q. Ma, X. Liu, and J. Jiang, "Rethinking 3D-CNN in hyperspectral image super-resolution," *Remote Sens.*, vol. 15, no. 10, p. 2574, May 2023.

[25] Q. Li, Q. Wang, and X. Li, "Exploring the relationship between 2D/3D convolution for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8693–8703, Oct. 2021.

[26] Q. Li, Q. Wang, and X. Li, "Mixed 2D/3D convolutional network for hyperspectral image super-resolution," *Remote Sens.*, vol. 12, no. 10, p. 1660, May 2020.

[27] Q. Li, Y. Yuan, X. Jia, and Q. Wang, "Dual-stage approach toward hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 31, pp. 7252–7263, 2022.

[28] D. Liu et al., "An efficient unfolding network with disentangled spatial-spectral representation for hyperspectral image super-resolution," *Inf. Fusion*, vol. 94, pp. 92–111, Jun. 2023.

[29] Y. Li, L. Zhang, C. Dingl, W. Wei, and Y. Zhang, "Single hyperspectral image super-resolution with grouped deep recursive residual network," in *Proc. IEEE 4th Int. Conf. Multimedia Big Data (BigMM)*, Sep. 2018, pp. 1–4.

[30] X. Wang, Q. Hu, J. Jiang, and J. Ma, "A group-based embedding learning and integration network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541416.

[31] Q. Wang, Q. Li, and X. Li, "Hyperspectral image superresolution using spectrum and feature context," *IEEE Trans. Ind. Electron.*, vol. 68, no. 11, pp. 11276–11285, Nov. 2021.

[32] D. Liu, J. Li, and Q. Yuan, "A spectral grouping and attention-driven residual dense network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7711–7725, Sep. 2021.

[33] X. Wang, J. Ma, and J. Jiang, "Hyperspectral image superresolution via recurrent feedback embedding and spatial–spectral consistency regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503113.

[34] J. Jiang, H. Sun, X. Liu, and J. Ma, "Learning spatial–spectral prior for super-resolution of hyperspectral imagery," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1082–1096, 2020.

[35] S. Chen, L. Zhang, and L. Zhang, "Cross-scope spatial–spectral information aggregation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 5878–5891, 2024.

[36] K. Wang, X. Liao, J. Li, D. Meng, and Y. Wang, "Hyperspectral image super-resolution via knowledge-driven deep unrolling and transformer embedded convolutional recurrent neural network," *IEEE Trans. Image Process.*, vol. 32, pp. 4581–4594, 2023.

[37] Q. Hu, X. Wang, J. Jiang, X.-P. Zhang, and J. Ma, "Exploring the spectral prior for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 5260–5272, 2024.

[38] X. Wang, J. Ma, J. Jiang, and X.-P. Zhang, "Dilated projection correction network based on autoencoder for hyperspectral image super-resolution," *Neural Netw.*, vol. 146, pp. 107–119, Feb. 2022.

[39] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[40] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.

[41] T. Uezato, M. Fauvel, and N. Dobigeon, "Hyperspectral image unmixing with LiDAR data-aided spatial regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4098–4108, Jul. 2018.

[42] X.-J. Mao, C. Shen, and Y.-B. Yang, "Image restoration using very deep convolutional encoder–decoder networks with symmetric skip connections," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 2810–2818.

[43] Z. Hui, X. Wang, and X. Gao, "Fast and accurate single image super-resolution via information distillation network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 723–731.

[44] X. Zhang et al., "Accurate and efficient image super-resolution via global-local adjusting dense network," *IEEE Trans. Multimedia*, vol. 23, pp. 1924–1937, 2021.

[45] T. Geng, X.-Y. Liu, X. Wang, and G. Sun, "Deep shearlet residual learning network for single image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 4129–4142, 2021.

[46] Y. Zhang, K. Li, K. Li, B. Zhong, and Y. Fu, "Residual non-local attention networks for image restoration," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–13.

[47] J. Liu, W. Zhang, Y. Tang, J. Tang, and G. Wu, "Residual feature aggregation network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 2356–2365.

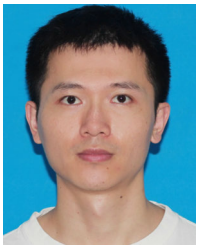
[48] Z. Zhu, J. Hou, J. Chen, H. Zeng, and J. Zhou, "Hyperspectral image super-resolution via deep progressive zero-centric residual learning," *IEEE Trans. Image Process.*, vol. 30, pp. 1423–1438, 2021.

[49] D. Liu, X. Wang, R. Han, N. Bai, J. Hou, and S. Pang, "CTE-Net: Contextual texture enhancement network for image super-resolution," *IEEE Trans. Multimedia*, vol. 26, pp. 8000–8013, 2024.

[50] Z. Du, D. Liu, J. Liu, J. Tang, G. Wu, and L. Fu, "Fast and memory-efficient network towards efficient image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 852–861.

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[51] H. Zhao, X. Kong, J. He, Y. Qiao, and C. Dong, "Efficient image super-resolution using pixel attention," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 56–72.
[52] T. Dai, J. Cai, Y. Zhang, S.-T. Xia, and L. Zhang, "Second-order attention network for single image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11057–11066.
[53] B. Niu et al., "Single image super-resolution via a holistic attention network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 191–207.
[54] B. Xia, Y. Hang, Y. Tian, W. Yang, Q. Liao, and J. Zhou, "Efficient nonlocal contrastive attention for image super-resolution," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 3, pp. 2759–2767.
[55] Y. Mei, Y. Fan, Y. Zhou, L. Huang, T. S. Huang, and H. Shi, "Image super-resolution with cross-scale non-local attention and exhaustive self-exemplars mining," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5689–5698.
[56] Y. Mei, Y. Fan, and Y. Zhou, "Image super-resolution with non-local sparse attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3516–3525.
[57] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.
[58] P. Ghosh, S. K. Roy, B. Koirala, B. Rasti, and P. Scheunders, "Hyperspectral unmixing using transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535116.
[59] W. Gao, Y. Zhang, Y. Akoudad, and J. Chen, "MSSF-Net: A multimodal spectral–spatial feature fusion network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511515.
[60] N. Yokoya and A. Iwasaki, "*Airborne hyperspectral data over Chikusei*," Dept. Space Appl. Lab., Univ. Tokyo, Tokyo, Japan, Tech. SAL2016-05-27, 2016.
[61] X. Huang and L. Zhang, "A comparative study of spatial approaches for urban mapping using hyperspectral ROSIS images over Pavia city, northern Italy," *Int. J. Remote Sens.*, vol. 30, no. 12, pp. 3205–3221, Jun. 2009.



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